

Linear Regression: Fuel Economy

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Case study

Set up libraries

```
pacman::p_load(  
  tidyverse,  
  tidymodels,  
  AppliedPredictiveModeling,  
  skimr,  
  janitor,  
  corrplot,  
  vip  
)
```

Read in data

```
data(FuelEconomy, package = "AppliedPredictiveModeling")
```

Pre-processing

Convert to tibble

```
cars2010 <- as_tibble(cars2010)
```

Clean names and select predictors

```
cars2010 <- clean_names(cars2010)
cars2010 <-
  cars2010 %>%
  select(
    eng_displ,
    num_cyl,
    FE = fe,
    num_gears,
    drive_desc
  )
cars2010
```

```
## # A tibble: 1,107 x 5
##   eng_displ num_cyl   FE num_gears drive_desc
##   <dbl>    <int> <dbl>    <int> <fct>
## 1     4.7      8  28.0      6 TwoWheelDriveRear
## 2     4.7      8  25.6      6 TwoWheelDriveRear
## 3     4.2      8  26.8      6 AllWheelDrive
## 4     4.2      8  25.0      6 AllWheelDrive
## 5     5.2     10  24.8      6 AllWheelDrive
## 6     5.2     10  23.9      6 AllWheelDrive
## 7      2      4  39.7      6 AllWheelDrive
## 8      6     12  24.4      6 AllWheelDrive
## 9      3      6  39.7      6 TwoWheelDriveRear
## 10     3      6  38.8      6 TwoWheelDriveRear
## # i 1,097 more rows
```

Recode the drive predictor

```
cars2010 %>%
  count(drive_desc)
```

```
## # A tibble: 5 x 2
##   drive_desc      n
##   <fct>        <int>
## 1 AllWheelDrive    205
## 2 FourWheelDrive   159
## 3 ParttimeFourWheelDrive  11
## 4 TwoWheelDriveFront  382
## 5 TwoWheelDriveRear  350
```

```
cars2010<-
  cars2010 %>%
  mutate(
```

```

drive = case_when(
  str_detect(drive_desc, "Front") ~ "front",
  str_detect(drive_desc, "Rear") ~ "rear",
  TRUE ~ "4WD"
))
cars2010 %>%
  count(drive_desc, drive)

```

```

## # A tibble: 5 x 3
##   drive_desc      drive      n
##   <fct>          <chr> <int>
## 1 AllWheelDrive    4WD    205
## 2 FourWheelDrive   4WD    159
## 3 ParttimeFourWheelDrive 4WD     11
## 4 TwoWheelDriveFront front   382
## 5 TwoWheelDriveRear rear    350

```

```

cars2010 <-
  cars2010 %>%
    select(-drive_desc)
cars2010

```

```

## # A tibble: 1,107 x 5
##   eng_displ num_cyl   FE num_gears drive
##   <dbl>    <int> <dbl>    <int> <chr>
## 1     4.7      8  28.0      6 rear
## 2     4.7      8  25.6      6 rear
## 3     4.2      8  26.8      6 4WD
## 4     4.2      8  25.0      6 4WD
## 5     5.2     10  24.8      6 4WD
## 6     5.2     10  23.9      6 4WD
## 7      2       4  39.7      6 4WD
## 8      6     12  24.4      6 4WD
## 9      3      6  39.7      6 rear
## 10     3      6  38.8      6 rear
## # i 1,097 more rows

```

Convert character into factor

```

cars2010 <-
  cars2010 %>%
    mutate_if(
      is.character, factor
    )
cars2010

```

```

## # A tibble: 1,107 x 5
##   eng_displ num_cyl   FE num_gears drive
##   <dbl>    <int> <dbl>    <int> <fct>
## 1     4.7      8  28.0      6 rear

```

```
## 2      4.7      8 25.6      6 rear
## 3      4.2      8 26.8      6 4WD
## 4      4.2      8 25.0      6 4WD
## 5      5.2     10 24.8      6 4WD
## 6      5.2     10 23.9      6 4WD
## 7      2       4 39.7      6 4WD
## 8      6      12 24.4      6 4WD
## 9      3       6 39.7      6 rear
## 10     3       6 38.8      6 rear
## # i 1,097 more rows
```

Split into training and testing

```
set.seed(2020)
car_split <- initial_split(cars2010, strata = FE)
car_train <- training(car_split)
car_test <- testing(car_split)
```

EDA

Overview

```
skim(car_train)
```

Table 1: Data summary

Name	car_train
Number of rows	829
Number of columns	5
Column type frequency:	
factor	1
numeric	4
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
drive	0	1	FALSE	3	fro: 287, 4WD: 279, rea: 263

Variable type: numeric

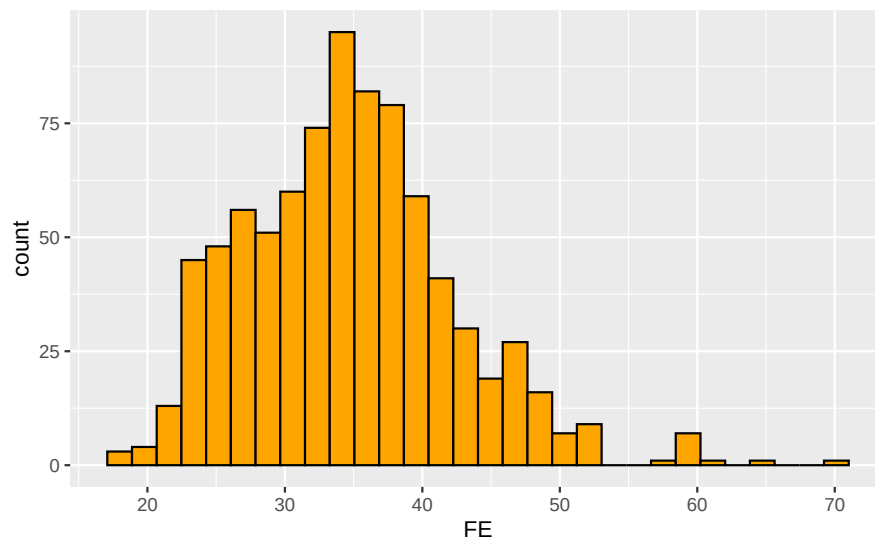
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
eng_displ	0	1	3.51	1.33	1.0	2.40	3.5	4.3	8.40	
num_cyl	0	1	5.99	1.96	2.0	4.00	6.0	8.0	16.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
FE	0	1	34.71	7.61	17.5	29.14	34.5	39.2	69.64	
num_gears	0	1	5.26	1.40	1.0	5.00	6.0	6.0	8.00	

Look at Fuel efficiency

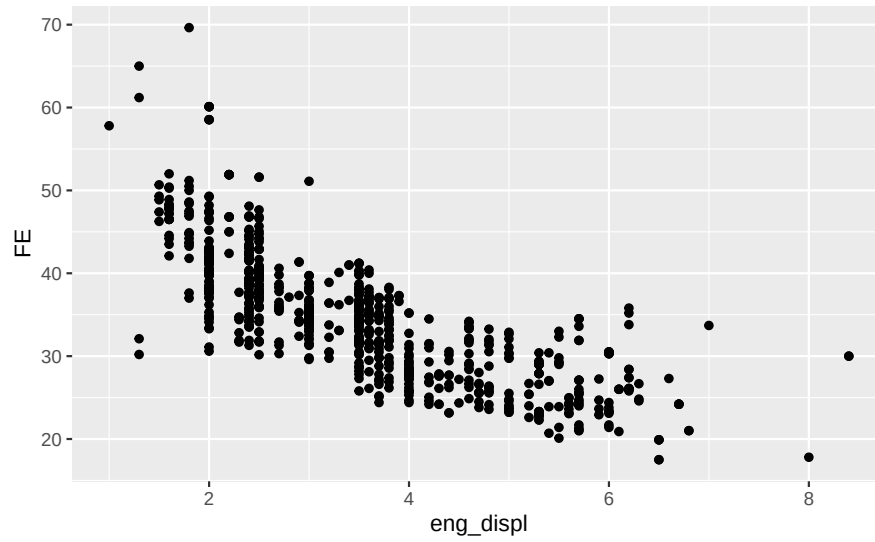
```
car_train %>%
  ggplot(aes(FE)) +
  geom_histogram(col = "black", fill = "orange")
```

`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.

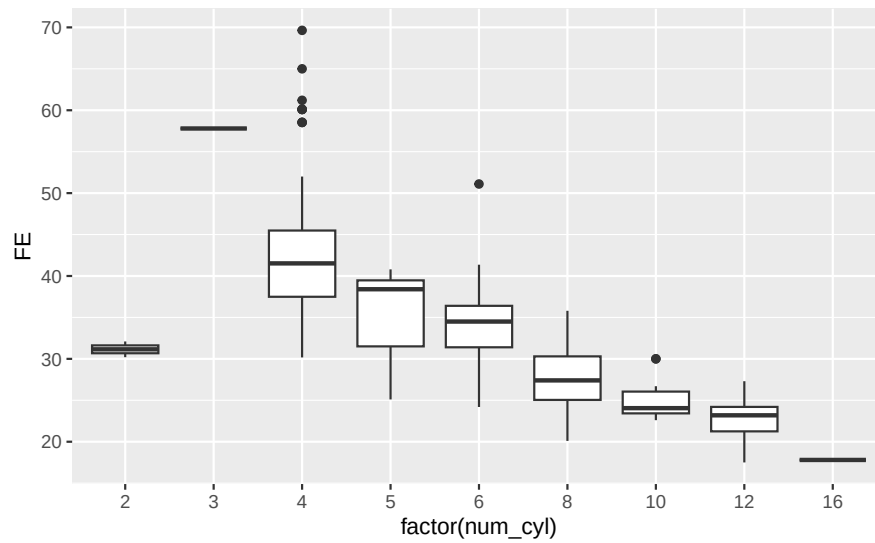


Bivariate relationships

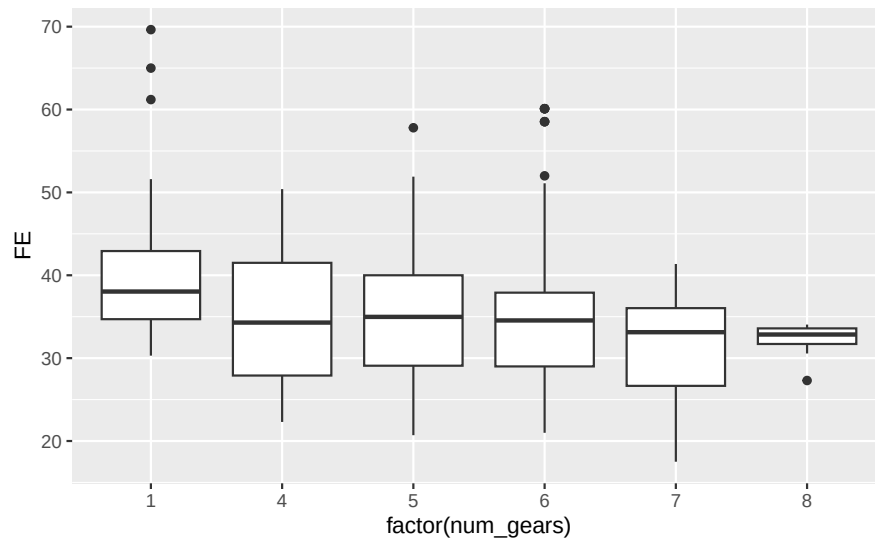
```
car_train %>%
  ggplot(aes(eng_displ, FE)) +
  geom_point()
```



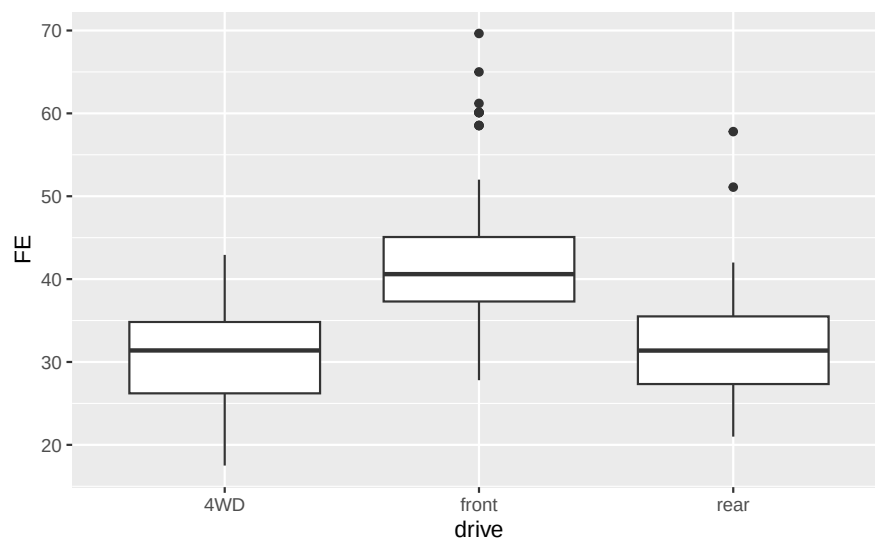
```
car_train %>%
  ggplot(aes(factor(num_cyl), FE)) +
  geom_boxplot()
```



```
car_train %>%
  ggplot(aes(factor(num_gears), FE)) +
  geom_boxplot()
```

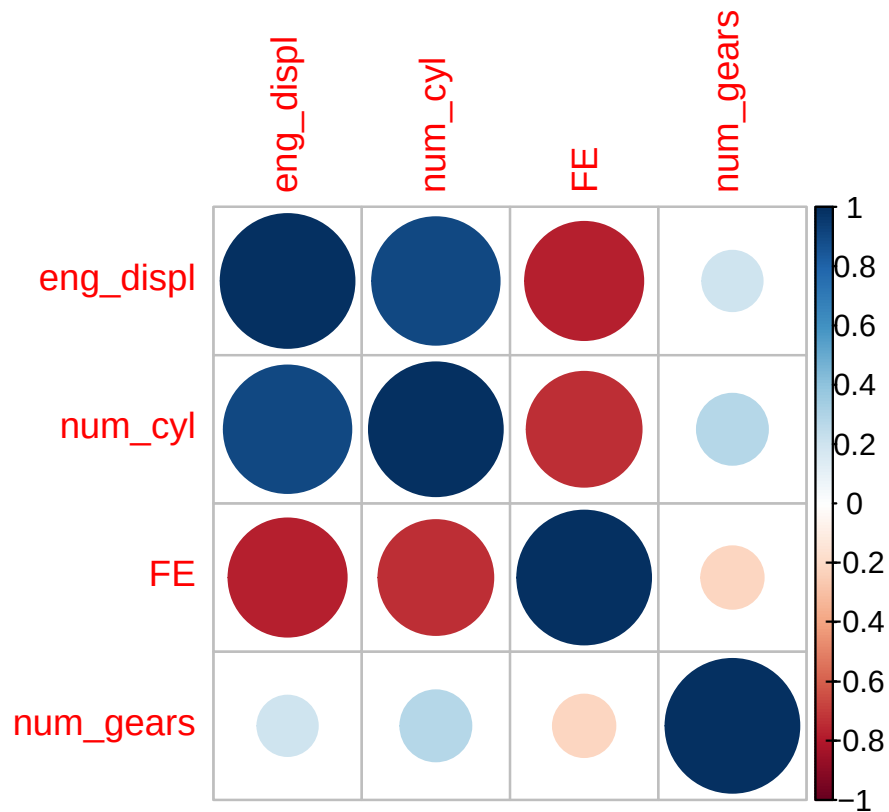


```
car_train %>%
  ggplot(aes(drive, FE)) + geom_boxplot()
```



Relationship between predictors

```
car_train %>%
  select_if(is.numeric) %>%
  cor() %>%
  corrplot()
```



Preprocessing with recipes

```
car_recipe <-
  recipe(FE ~ ., data = car_train) %>%
    step_poly(eng_displ, degree = 2, options = list(raw = TRUE))
car_recipe
```

```
##
```

```
## -- Recipe -----
```

```
##
```

```
## -- Inputs
```

```
## Number of variables by role
```

```
## outcome: 1
## predictor: 4
```

```
##
```

```
## -- Operations
```



```
## * Orthogonal polynomials on: eng_displ

car_recipe <- car_recipe %>% prep()
car_recipe

##

## -- Recipe -----

##

## -- Inputs

## Number of variables by role

## outcome: 1
## predictor: 4

##

## -- Training information

## Training data contained 829 data points and no incomplete rows.

##

## -- Operations

## * Orthogonal polynomials on: eng_displ | Trained

car_train <- car_recipe %>% juice()
car_train

## # A tibble: 829 x 6
##   num_cyl num_gears drive    FE eng_displ_poly_1 eng_displ_poly_2
##   <int>    <int> <fct> <dbl>          <dbl>          <dbl>
## 1      8      6 rear  28.0          4.7          22.1
## 2      8      6 rear  25.6          4.7          22.1
## 3      8      6 4WD  26.8          4.2          17.6
## 4      8      6 4WD  25.0          4.2          17.6
## 5     12      6 4WD  24.4          6           36
## 6     16      7 4WD  17.8          8           64
## 7      8      7 rear  24.3          4.5          20.2
## 8     12      6 rear  21.0          5.7          32.5
## 9     12      6 rear  21.1          5.7          32.5
## 10     10      6 4WD  25.4          5.2          27.0
## # i 819 more rows
```

```
car_test <- car_recipe %>% bake(car_test)
car_test
```

```
## # A tibble: 278 x 6
##   num_cyl num_gears drive    FE eng_displ_poly_1 eng_displ_poly_2
##   <int>    <int> <fct> <dbl>          <dbl>          <dbl>
## 1      10         6 4WD    24.8            5.2            27.0
## 2      10         6 4WD    23.9            5.2            27.0
## 3       6         6 rear    39.7             3              9
## 4       6         6 rear    38.8             3              9
## 5       6         7 rear    33.6             3              9
## 6       8         6 rear    27.1            6.2           38.4
## 7       8         6 rear    34.3            6.2           38.4
## 8       6         7 rear    36.2             3.5            12.2
## 9       8         7 rear    30.8             5.5            30.2
## 10      3         5 rear    57.8             1              1
## # i 268 more rows
```

Fit model

```
car_lm <-
  linear_reg() %>%
  set_engine("lm") %>%
  fit(FE ~ ., data = car_train)
car_lm
```

```
## parsnip model object
##
##
## Call:
## stats::lm(formula = FE ~ ., data = data)
##
## Coefficients:
##      (Intercept)          num_cyl          num_gears          drivefront
##          54.90897         -0.55180           0.06216           5.25074
##      driverear eng_displ_poly_1 eng_displ_poly_2
##          2.24564          -8.28921           0.66325
```

Use model

Model fit

```
car_train %>%
  add_column(
    predict(car_lm, new_data = car_train)
  ) %>%
  yardstick::rmse(truth = FE, estimate = .pred)
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 rmse    standard       3.94
```

```
car_test %>%
  add_column(
    predict(car_lm, new_data = car_test)
  ) %>%
  yardstick::rmse(truth = FE, estimate = .pred)
```

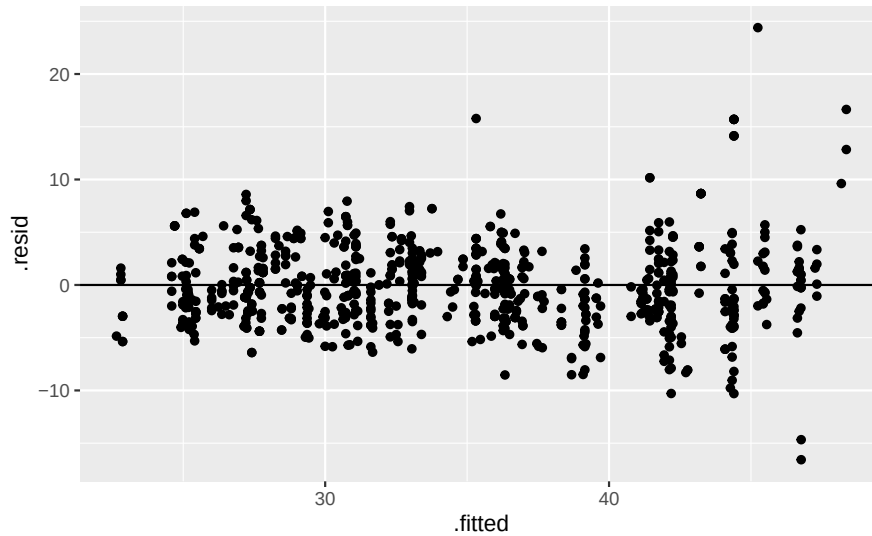
```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 rmse    standard       3.38
```

Check assumptions

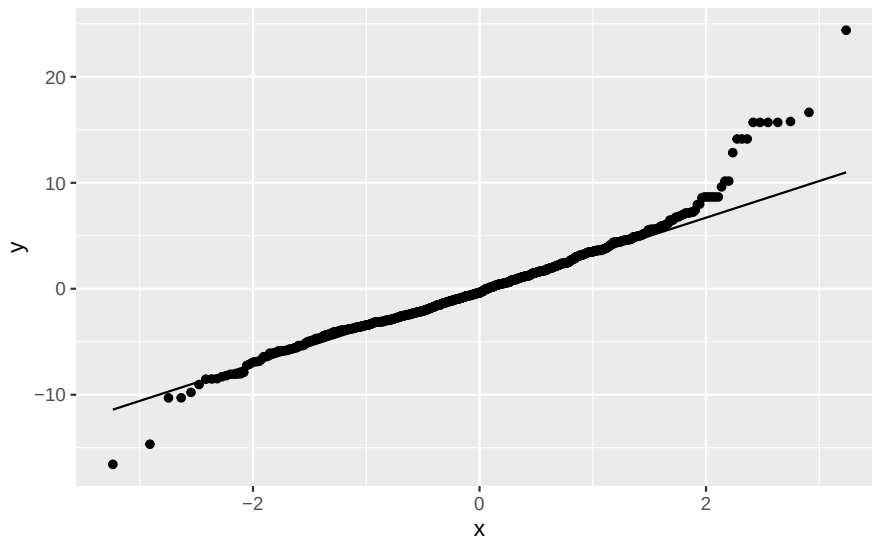
```
car_res <- augment(car_lm$fit)
car_res
```

```
## # A tibble: 829 x 12
##       FE num_cyl num_gears drive eng_displ_poly_1 eng_displ_poly_2 .fitted
##   <dbl>   <int>   <int> <fct>   <dbl>         <dbl>   <dbl>
## 1  28.0     8       6 rear     4.7          22.1   28.8
## 2  25.6     8       6 rear     4.7          22.1   28.8
## 3  26.8     8       6 4WD     4.2          17.6   27.8
## 4  25.0     8       6 4WD     4.2          17.6   27.8
## 5  24.4    12       6 4WD     6            36     22.8
## 6  17.8    16       7 4WD     8            64     22.6
## 7  24.3     8       7 rear     4.5          20.2   29.3
## 8  21.0    12       6 rear     5.7          32.5   25.2
## 9  21.1    12       6 rear     5.7          32.5   25.2
## 10 25.4    10       6 4WD     5.2          27.0   24.6
## # i 819 more rows
## # i 5 more variables: .resid <dbl>, .hat <dbl>, .sigma <dbl>, .cooksd <dbl>,
## #   .std.resid <dbl>
```

```
car_res %>%
  ggplot(aes(.fitted, .resid)) +
  geom_point() +
  geom_hline(yintercept = 0)
```



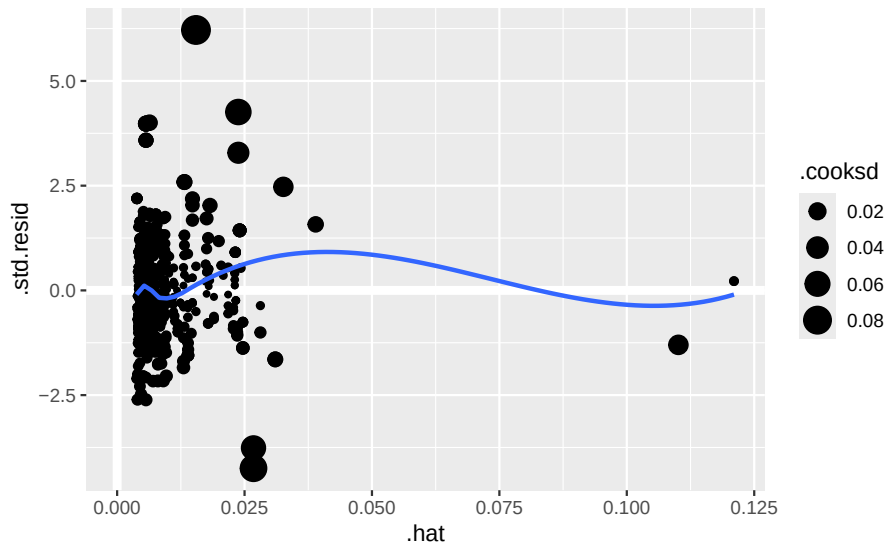
```
car_res %>%
  ggplot(aes(sample = .resid)) +
  stat_qq() +
  stat_qq_line()
```



```
car_res %>%
  ggplot(aes(.hat, .std.resid)) +
  geom_vline(size = 2, colour = "white", xintercept = 0) +
  geom_hline(size = 2, colour = "white", yintercept = 0) +
  geom_point(aes(size = .cooks)) + geom_smooth(se = FALSE)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



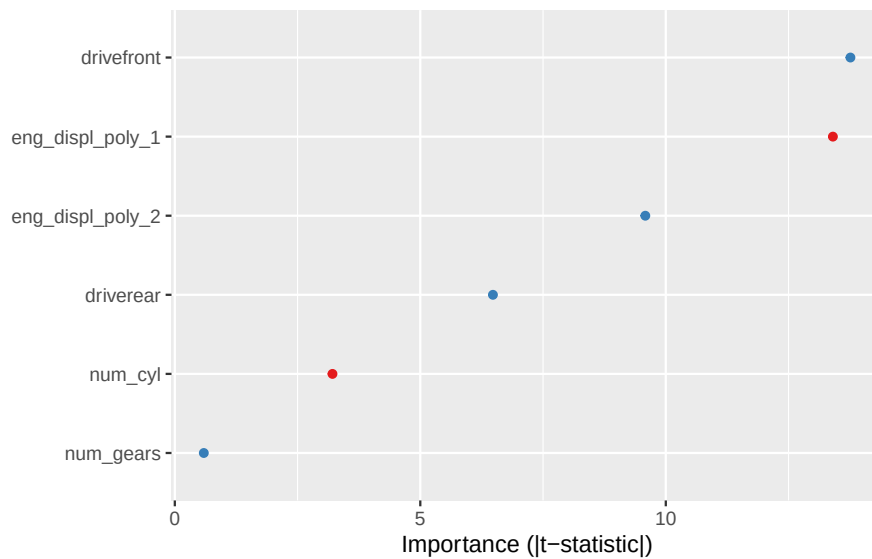
Variable importance

```
vi(car_lm)
```

```
## # A tibble: 6 x 3
##   Variable      Importance Sign
##   <chr>          <dbl> <chr>
## 1 drivefront      13.8  POS
## 2 eng_displ_poly_1 13.4  NEG
## 3 eng_displ_poly_2  9.58  POS
## 4 driverear       6.48  POS
## 5 num_cyl         3.21  NEG
## 6 num_gears        0.590  POS
```

```
vip(car_lm,
  include_type = TRUE,
  geom = 'point',
  mapping = aes_string(col = "Sign"), size = 5) +
  scale_color_brewer(palette = "Set1")
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Inference

```
tidy(car_lm$fit)
```

```
## # A tibble: 7 x 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)    54.9      1.24     44.2 2.68e-219
## 2 num_cyl       -0.552    0.172    -3.21 1.37e- 3
## 3 num_gears      0.0622   0.105     0.590 5.55e- 1
## 4 drivefront     5.25     0.382    13.8 6.34e- 39
## 5 driverear      2.25     0.347     6.48 1.59e- 10
## 6 eng_displ_poly_1 -8.29    0.618   -13.4 3.40e- 37
## 7 eng_displ_poly_2  0.663   0.0692    9.58 1.08e- 20
```

```
tidy(car_lm$fit) %>%
  arrange(p.value)
```

```
## # A tibble: 7 x 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 drivefront     5.25     0.382    13.8 6.34e- 39
## 2 eng_displ_poly_1 -8.29    0.618   -13.4 3.40e- 37
## 3 eng_displ_poly_2  0.663   0.0692    9.58 1.08e- 20
## 4 driverear      2.25     0.347     6.48 1.59e- 10
## 5 num_cyl       -0.552    0.172    -3.21 1.37e- 3
## 6 num_gears      0.0622   0.105     0.590 5.55e- 1
```

Prediction

```
new_data <- tibble(
  num_cyl = 4,
  num_gears = 6,
  eng_displ = 5,
  drive = "front"
)
new_data
```

```
## # A tibble: 1 x 4
##   num_cyl num_gears eng_displ drive
##   <dbl>    <dbl>    <dbl> <chr>
## 1      4        6        5 front
```

```
new_data <- bake(car_recipe, new_data = new_data)
```

```
## Warning: ! There was 1 column that was a factor when the recipe was prepped:
## * `drive`
## i This may cause errors when processing new data.
```

```
new_data
```

```
## # A tibble: 1 x 5
##   num_cyl num_gears drive eng_displ_poly_1 eng_displ_poly_2
##   <dbl>    <dbl> <fct>          <dbl>          <dbl>
## 1      4        6 front              5              25
```

```
predict(car_lm, new_data = new_data)
```

```
## # A tibble: 1 x 1
##   .pred
##   <dbl>
## 1  33.5
```

```
predict(car_lm, new_data = new_data, type = "pred_int")
```

```
## # A tibble: 1 x 2
##   .pred_lower .pred_upper
##   <dbl>    <dbl>
## 1    25.5    41.4
```